

Name _____

Section _____

MTH:2310 DISCRETE MATH
MIDTERM 3

Complete all of the following problems being as detailed as you can be. Remember to fully define new variables/symbols. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit. Do your best to make your solutions organized and written legibly.

Handwriting. (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 2 points for arithmetic or notation errors.

You will be penalized up to 2 points for disorganized or hard-to-read solutions.

Problem 1. (8 points)

Determine whether 119 is prime using Trial Division. You may use $\sqrt{119} \approx 10.909$.

Problem 2. (10 points)

Evaluate the following expression.

$$\left((6^2 \cdot 7) \pmod{5} + (-3) \pmod{7} \right) \pmod{3}$$

Problem 3. (8 points)

Let $f : \mathbb{N} \rightarrow \mathbb{Q}$ be the function defined as $f(0) = 2$, and $f(n+1) = \left(\frac{f(n)}{n+1}\right)^{n-1}$ whenever $n \geq 0$. Find the value of $f(1)$, $f(2)$.

Problem 4. (6 points)

Find the multiplicative inverse of 3 modulo 5, if it exists. If not, clearly explain why.

Name _____

Section _____

Problem 5. (14 points)

Prove that for any positive integer n , $(1 \cdot 2) + (2 \cdot 3) + \cdots + (n \cdot (n + 1)) = \frac{n(n+1)(n+2)}{3}$.

Name _____

Section _____

Problem 7. (6 points)

State the general form of the solution to the recurrence relation with the following characteristic equation.

$$(r + 1)^2(r + 5)^3 = 0$$

Problem 8. (12 points)

Solve the system of linear congruences. You may use either the construction method or the substitution method. Clearly state which method you are using and show every step to find the solution. **No other method will be graded.**

$$\begin{cases} x \equiv 1 \pmod{2} \\ x \equiv 4 \pmod{5} \end{cases}$$

Problem 9. (16 points)

- (a) (4 points) Find the prime factorization of 220 and 1400.
- (b) (2 points) Find the greatest common divisor of 220 and 1400 using their prime factorizations. You do not need to simplify your answer.
- (c) (10 points) Use the Euclidean algorithm to find the greatest common divisor of 220 and 1400. **Show every step of the algorithm.**